

PREMA M

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OBJECTIVE

Experienced embedded software professional with 1.7 years of expertise in firmware development. Proficient in communication protocols (I2C, SPI, UART, EEPROM) and modularized coding. Skilled in Arduino and code composer studio. Strong ability in testing and debugging.

EXPERIENCE

Tantragyaan Solutions (2023 March - 2024 October)

Product Engineer - Firmware

- Develop firmware for sensors such as Temperature sensor (**TMP117**), **BME** and **BMP** sensors.
- Specializing in communication protocols such as **I2C**, **SPI**, **UART** and **EEPROM**.
- Implement **modularized coding** practices to ensure maintainability and code quality.
- Conduct unit testing, module testing and integration testing to validate software functionality and reliability.
- Worked on **interrupt** handling functionality.
- Debug issues during **board bring-up**.
- Maintain through documentation of code for future reference.

TECHNICAL SKILLS

- Firmware development for different sensors using Embedded C and C++
- Interfaces : I2C, SPI, UART, EEPROM, GPIO
- Communication protocols : UDP, TCP
- Modularized coding
- IDE : Arduino, Code composer studio
- Documentation : Doxygen

PROJECTS

● SVASTHYA SMART ACTIVITY TRACKER

- Svasthya is a Bio-informatics platform for remote health discovery and delivery.
- Measures the bio-vitals at real-time and brings the doctor, patient and family members on a collaborative platform for viewing data from anywhere in the world.
- **Platform** : ARDUINO IDE
- **Key Responsibilities:**
 - Added beat detector algorithm to measure the heart rate.
 - Implemented the feature of dynamically adjusting the LED current to read the SPO2 data.

- Bring up Step counter, Activity classifier and Stability detector.
- Implemented History data packet creation and Live data packet creation.
- Developed the alarm wake up and Accelerometer wake up functionality by attaching the interrupts.
- Worked on writing data to the SPI Flash, reading data from it, clearing the Flash memory and Flash memory mapping.

● **Skills :** Embedded C++, I2C, SPI, UART, RTC, SPI Flash

● **GSR - Galvanic Skin Response**

● **Platform :** ARDUINO IDE

● **Key Responsibilities :**

- GSR Sensor data collection at the rate of 100 samples per second.
- Averaged every five consecutive raw sensor data readings and stored the result in a buffer to reduce the data collection rate from 100 sps to 20 sps.
- Implemented code for GSR Packet Creation by wrapping the averaged GSR data and current timestamp.
- Transferred the GSR Packets to the application through BLE connection for every 200 milliseconds at the rate of 20 samples per seconds.

● **Skills :** Embedded C++, UART

● **SVASTHYA CLINICAL TRIALS**

● **Platform :** ARDUINO IDE

● **Key Responsibilities :**

- Developed the wake_up functionality to wake_up Svasthya devices for every 30 minutes.
- The device transfers all the vital values to the application through BLE connection for 3 minutes.
- After data transmission, the device sets an alarm to next 30 minutes cycle and enters sleep mode.
- Macro functionality is implemented for the alarm wake_up, data transmission timing, and ECG data capture.

● **Skills :** Embedded C++, UART, RTC

● **GENERIC COMMUNICATION MANAGER**

● **Platform :** Code Composer Studio

● **Key Responsibilities :**

- Developed a software to establish client-server connection, read and write functionality with the server using UDP protocol.
- Implemented the modularized coding for clarity and maintainability.
- Documentation was maintained using Doxygen to ensure proper code documentation and facilitate easier understanding and future development.

● **Skills :** Embedded C, UDP Protocol, Modularized coding, Doxygen

EDUCATION

RNS INSTITUTE OF TECHNOLOGY, BENGALURU

B.E (Information Science and Engineering) with the aggregate of 8.75
VTU, 2019 - 2023

GOVT. PU COLLEGE FOR BOYS, TIPTUR

XII Standard with the percentage of 91
2019

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge.

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